Examiner reports

Q1.

There were several good solutions to part (a), but some candidates ignored the instruction to use two equations of motion. Some candidates assumed that A would move up the slope and then ran into difficulties with the negative signs that were involved. Many candidates found part (b) difficult. It was very common for them to try to use the acceleration from part (a) instead of setting the acceleration equal to zero. Interestingly, some candidates who made little or no progress with part (a) did complete part (b) successfully.

Q2.

A lot of the candidates found this question very difficult. There were a number of good clear force diagrams, but there were many that included a friction force. Some also introduced a force acting down the slope. A few candidates did not make the directions of the forces clear. Many candidates were unable to resolve horizontally in part (b). The most common error was to resolve the 8 N force instead of the reaction force, obtaining equations such as R = 8 cos 30º. In part (c), many candidates produced results such as R = mg cos θ, by resolving perpendicular to the slope and ignoring the 8 N force. A few simply stated that R = mg.

Q3.

Part (a) was usually done very well. However there were very few good responses to part (b). The most common error was to try to find the magnitude of the resultant of P and the friction, assuming that the friction force had magnitude 58.8 N. For many candidates their understanding of the friction inequality seemed poor. In part (c), the candidates tended to either find the correct value or to simply give the answer 16 N, which was the product of the mass and the acceleration. Many candidates did part (d) very well, but a few omitted the negative sign from the acceleration. Those candidates who had difficulties usually used an incorrect acceleration.

Q4.

There were many good responses to this question, with many candidates scoring full marks. Some candidates resolved horizontally and vertically, while others used a vector triangle approach combined with the sine rule. The most common error was to resolve incorrectly or use the wrong angles.

Q5.

There were a number of good responses to this question. Some parts caused the candidates a little more difficulty than the earlier questions. In part (a), some candidates included additional forces. Parts (b) and (c) were generally well done, the only significant problem was that some candidates used the wrong angle when finding the components of the weight and so ended up with their answers to parts (b) and (c) interchanged. This caused subsequent problems in part (d). However, most candidates made a good attempt at part (d). Part (e) was more challenging. Many candidates felt that the 14 kg would just slide down the slope. Some argued that the friction and weight would be proportional to the mass and so it would stay still. Quite a number repeated all the earlier calculations to obtain the same coefficient of friction and often concluded that the block would remain at rest.